



Unit 9 Essential Knowledge & Guided Notes

Formulas, Reactions, and Equations (Part 1 and 2)

Textbook: CK Chemistry Chapter 10 - The Mole, Chapters 11 - Chemical Reactions
<https://flexbooks.ck12.org/cbook/ck-12-chemistry-flexbook-2.0/>

Helpful Videos and Practice: Posted on CANVAS

Essential Knowledge:

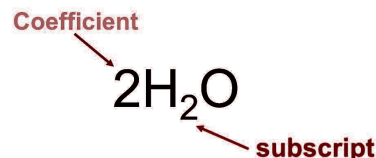
I. Percent Composition and Empirical/Molecular Formulas

- To calculate the % by mass of elements in a compound use this equation:

$$\% \text{ mass} = \frac{\text{mass of the element in the compound}}{\text{molar mass of the compound}} \times 100$$
- Definitions:
 - The empirical formula shows the simplest whole-number ratio in which the atoms of the elements are present in the compound.
 - The molecular formula shows the actual number of atoms of each element in one molecule of the substance.
- Given mass data, you can calculate the empirical and molecular formulas of compounds. Examples will be given in class.
- Hydrated compounds contain a specific number of water molecules embedded in the crystal for every unit as represented by a dot in the formula. The number of water molecules is added to the name using a prefix.

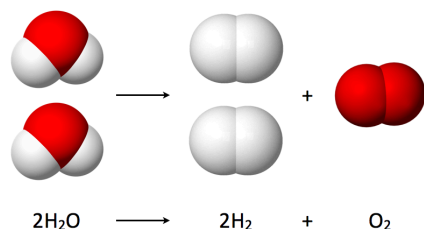
II. Balancing Chemical Equations

- In a chemical equation, the reactants are the starting materials and products are the ending materials in a chemical reaction.
- Chemical Reactions are rearrangements of atoms and mass is not lost or gained. This is the Law of Conservation of Matter
- In a chemical equation, coefficients are used so that the same number and type of atom appear on both the reactant and product sides.



Coefficients are used to balance equations.
 NEVER change or add subscripts!

REACTANT → PRODUCTS
 MASS OF REACTANTS = MASS OF PRODUCTS



4 atoms of H = 4 atoms of H
 2 atoms of O = 2 atoms of O

Balancing Means:

Conservation of Matter Means:

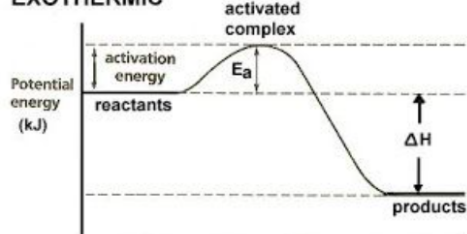
2 molecules of H_2O → 2 molecules H_2 + 1 molecule of O_2
 $2(18 \text{ amu}) = 2(2 \text{ amu}) + 1(32 \text{ amu})$
 2 moles of H_2O → 2 moles of H_2 + 1 mole of O_2
 $2(18 \text{ g}) = 2(2 \text{ g}) + 1(32 \text{ g})$

III. Classifying and Writing Chemical Reaction Types

- There are 6 types of chemical reactions: synthesis, decomposition, single replacement, double replacement, neutralization, and/or combustion.
 - synthesis ($\text{A} + \text{B} \rightarrow \text{AB}$)
 - decomposition ($\text{BC} \rightarrow \text{B} + \text{C}$)
 - single replacement ($\text{A} + \text{BC} \rightarrow \text{B} + \text{AC}$)
 - double replacement ($\text{AC} + \text{BD} \rightarrow \text{AD} + \text{BC}$)
 - neutralization ($\text{HX} + \text{MOH} \rightarrow \text{H}_2\text{O} + \text{MX}$)
 - combustion ($\text{C}_x\text{H}_y + \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$)

- Using knowledge of these patterns, the products of the types of chemical reaction identified above can be predicted (Practice will be given in your homework)
- An **activity or reactivity series** can be used to determine whether a single replacement reaction will occur. Elements higher in the activity series can replace elements lower in the activity series in an ionic compound, but not the other way around. (See Reference p.3)
- A single replacement reaction is an oxidation-reduction reaction (redox reaction). In **oxidation**, electrons are lost and in **reduction** those electrons are gained. Oxidation numbers can be assigned to see this.
- Many ionic compounds will dissolve in water so we say they are **soluble**. When these compounds are mixed with water they dissolve, and we describe them as **aqueous (aq)**. There are many ionic compounds that do not dissolve in water, though. These are described as **insoluble**. An insoluble substance simply remains in the **solid state (s)** when added to water. Solubility rules can be used to predict the **precipitate** (insoluble product) in a double replacement reaction. (See Reference p.3)
- A **spectator ion** is an ion that is present during a reaction but does not take part in the reaction. A **net ionic equation** is an equation that only shows the ions that undergo changes during a chemical reaction. (Spectator ions are omitted from net ionic equations.)
- Reactions can be **exothermic** (releasing heat) or **endothermic** (using heat)
 - Enthalpy(H)** is a measure of the energy that is released or absorbed when bonds are broken and formed during a reaction.
 - When bonds are formed, energy is released
 - When bonds are broken, energy must be absorbed to break them.
 - $\Delta H = H_{\text{products}} - H_{\text{reactants}}$

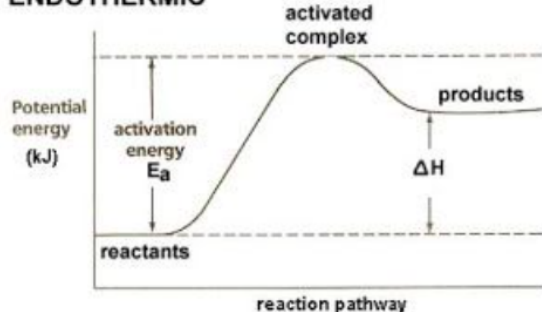
EXOTHERMIC



e. If the products of a chemical reaction have stronger bonds than the reactants, then more energy is released in making of product bonds than was put in to break the reactant bonds. The reaction is exothermic ΔH is negative. Energy is a product.



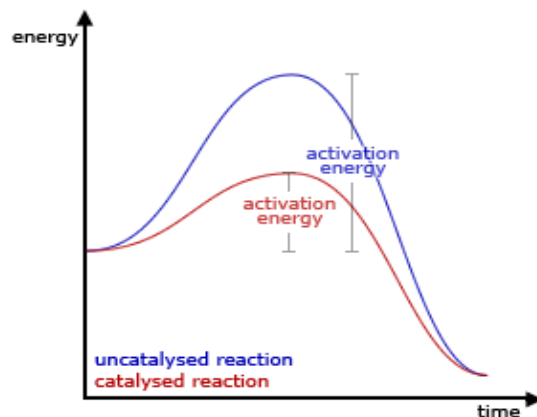
ENDOTHERMIC



f. If the products of a chemical reaction have weaker bonds than the reactants, then less energy is released in making of product bonds than was put into breaking the reactant bonds. The reaction is endothermic. ΔH is positive. Energy is a reactant.



- The **Activation Energy (E_A)** is the energy required to break the bonds and start the reaction. It is always endothermic.
- A **catalyst** speeds up a reaction by lowering the activation energy.



Reference:

K	Potassium		Most reactive
Na	Sodium		
Ca	Calcium		
Mg	Magnesium		
Al	Aluminium		
C	Carbon		
Zn	Zinc		
Fe	Iron		
Sn	Tin		
Pb	Lead		
H	Hydrogen		Least reactive
Cu	Copper		
Ag	Silver		
Au	Gold		
Pt	Platinum		
C	H	added for comparison	

Reactivity Series of Metals

Compounds Containing the Following Ions Are Generally Soluble	Exceptions
Group I and NH_4^+	none
NO_3^- and $\text{C}_2\text{H}_3\text{O}_2^-$	none
Cl^- , Br^- , and I^-	when paired with Ag^+ , Hg_2^{2+} , or Pb^{2+} (insoluble)
SO_4^{2-}	when paired with Sr^{2+} , Ba^{2+} , Pb^{2+} , Ag^+ , or Ca^{2+} (insoluble)
Compounds Containing the Following Ions Are Generally Insoluble	Exceptions
S^{2-}	when paired with group I, NH_4^+ , Ca^{2+} , Sr^{2+} , or Ba^{2+} (soluble)
OH^-	when paired with group I and NH_4^+ (soluble) or when paired with Ca^{2+} , Sr^{2+} , or Ba^{2+} (slightly soluble)
CO_3^{2-} and PO_4^{3-}	when paired with group I or NH_4^+

U9 Lesson #1 - **Percent Composition by Mass**

- The molar mass is the sum of the individual atomic masses of each atom in the compound.
- Find the molar mass of Fe_2O_3 (Round all molar masses to the hundredth place)
- $\% \text{ mass} = \frac{\text{mass of the element in the compound}}{\text{molar mass of the compound}} \times 100$
- Chemical Analysis Problem: Find % mass of Iron in Iron (III) oxide (rust), Fe_2O_3 .
Mass of the compound =
Mass of the iron in the compound =

% mass =
- Iron can be reclaimed from iron (III) oxide using a _____.
- Chemical Analysis Problem: Using the percent found, what mass of iron can be recovered from 7.8 grams of iron(III) oxide?
- Chemical Analysis Problem: Find the % mass of hydrogen in H_2O ?
Mass of the compound =
Mass of the hydrogen in the compound =

% mass =
- Gypsum is a hydrated salt. It contains a specific number of water molecules embedded in the crystal for every unit of calcium sulfate.
Formula _____ Name _____

9. What does the dot mean in the formula?

10. Memorize these prefixes which are the same as covalent compounds: **mono (1), di (2), tri (3), tetra (4), penta (5), hexa (6), hepta (7), octa (8), nona (9) and deca (10)**

11. Practice Naming:

Name	Formula
lithium bromide trihydrate	
	$\text{AlCl}_3 \cdot 6 \text{H}_2\text{O}$
magnesium sulfate heptahydrate	
	$\text{Ba}(\text{OH})_2 \cdot 8 \text{H}_2\text{O}$

12. Anhydrous means **without water**.

a. Name: $\text{CaSO}_4 \cdot 2\text{H}_2\text{O} =$ _____

b. Name: $\text{CaSO}_4 =$ _____

13. Chemical Analysis Problem: Find the % mass of water in $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$

Mass of the compound =

Mass of the water in the compound =

% mass =

14. Chemical Analysis Problem

Name: $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ - Sodium carbonate _____

Find the % mass of water in $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$

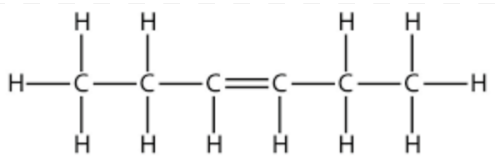
Mass of the compound =

Mass of the water in the compound =

% mass =

U9 Lesson #2 – Empirical and Molecular Formulas

1. The **molecular formula** shows the ratio of the elements in a compound.



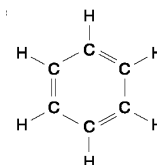
2. The **empirical formula** is the lowest whole number ratio of the molecular formula.

3. The empirical formula for the molecular formula $\text{C}_6\text{H}_{12} =$ _____

4. This is a molecule of the covalent compound benzene.

What is its molecular formula? _____

What is its empirical formula? _____



5. Molecular = $\text{C}_6\text{H}_{12}\text{O}_3$ Empirical = _____
 Molecular = CO_2 Empirical = _____
6. Chemical Analysis Problem: A compound has an empirical formula of CH_3 Its molar mass is 30 grams. What is its molecular formula? Follow these steps:
 - a. Find the mass of the empirical formula.
 - b. Determine the number of times it goes into the molar mass
 - c. Multiply the empirical formula by that number
 - d. Write molecular formula _____
7. Chemical Analysis Problem: A compound has an empirical formula of CH_2 and molar mass is 70 grams.
 - a. Find the mass of the empirical formula.
 - b. Determine the number of times it goes into the molar mass
 - c. Multiply the empirical formula by that number
 - d. Write molecular formula _____
8. Chemical Analysis Problem: What is the empirical formula of a compound that is 7.7% H and 92.3% C?
 - a. Pretend % is grams
 - b. Determine # of moles
 - c. Determine the lowest whole # ratio by division
 - d. Write the empirical formula:

9. Chemical Analysis Problem: What is the empirical formula of a compound that is 20.0% H and 80.0 % C by mass? What would the molecular formula be if the molar mass was 30.08 grams?
- Pretend % is grams
 - Determine # of moles
 - Determine the lowest whole # ratio by division
 - Write the empirical formula:
 - Then follow the steps to find the molecular formula from the Empirical formula and the molar mass. Look at #7

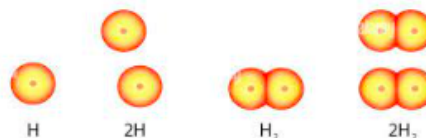
Molecular formula =

10. Chemical Analysis Problem with grams instead of %: Find the empirical formula of a compound containing 13.323g copper and 1.677 g oxygen.
- Determine # of moles
 - Determine the lowest whole # ratio by division
 - Write the empirical formula:

- d. Then follow the steps to find the molecular formula from the Empirical formula and the molar mass. Look at #7

U9 Lesson #3 Balancing Equations

- After watching the video clip in the lesson, answer these questions:
 - Can you fix George's (actor Steve Martin) problem? _____ How many packages of buns and how many packages of hot dogs does he need to buy to have an even ratio of hot dogs to buns? _____ hot dog packages: _____ bun packages
 - What is the math term that you used to figure out "a"? _____
 - If this food scenario was a chemical reaction, what would the reactant(s) be? _____ and _____ What would the product(s) be? _____
 - Look at the various examples for hydrogen. A coefficient (ex: 2H) is used to indicate the number of molecules or atoms. A subscript (ex: H₂) is used to indicate how many hydrogen are bonded in a molecule.

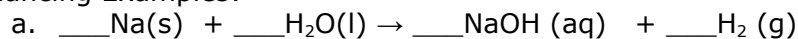


- Try to write a **balanced** equation describing this food situation using only the symbols: B and D (for buns and dogs). Use coefficients and subscripts as appropriate:



- The law of conservation of mass states that in a chemical reaction, matter can neither be created nor destroyed. Atoms are just rearranged.**
- Label the subscript and coefficient: $3\text{H}_2\text{O}$
- Draw structural formulas representing $3\text{H}_2\text{O}$:

5. Balancing Examples:



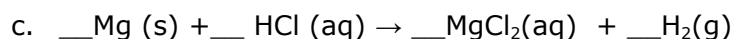
Check:

Reactants Atoms after Balancing	=	Product Atoms after Balancing



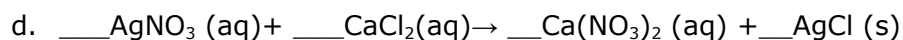
Check:

Reactants Atoms after Balancing	=	Product Atoms after Balancing



Check:

Reactants Atoms after Balancing	=	Product Atoms after Balancing



Hint: Treat polyatomic ions as one unit if they appear on both sides of the equation.

Check:

Reactants Atoms after Balancing	=	Product Atoms after Balancing

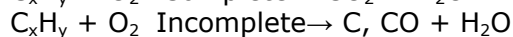
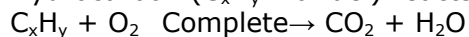
Now Try the pHet balancing practice on CANVAS!

Unit 9 Lesson #4 – Types of Reactions



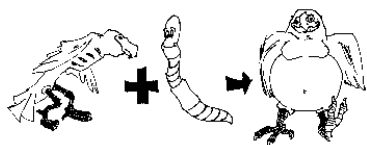
Type 1: Combustion:

Hydrocarbon (C_xH_y – a fuel) reacts with oxygen and burns

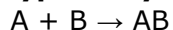


Word Equation:

Balanced Equation:



Type 2: Synthesis (or Combination!)



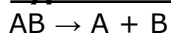
Two elements or compounds form a single compound

Word Equation:

Balanced Equation:



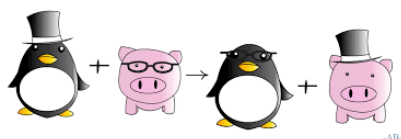
Type 3: Decomposition:



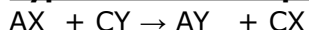
A single compound breaks up into two elements or compounds

Word Equation:

Balanced Equation:



Type 4: Double Replacement:

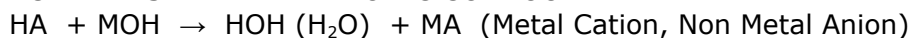


Two compounds $\rightarrow$ Two Compounds

Word Equation:

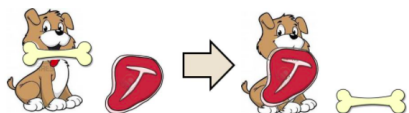
Balanced Equation:

Special Type of Double Replacement – Neutralization



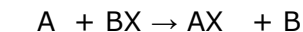
Word Equation Example:

Balanced Equation:



Type 5: Single Replacement:

(More Active Element replaces the less Active Element. Metals replace metals only. Non metals replace non metals only.)



compound + element $\rightarrow$ Compound + element

Word Equation Example:

Balanced Equation:

Word Equation Example:

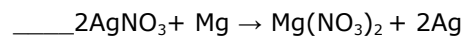
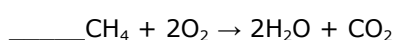
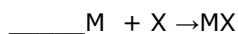
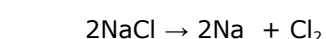
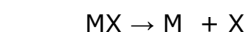
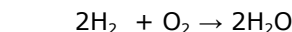
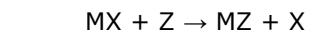
Balanced Equation:

K	Potassium	 Most reactive
Na	Sodium	
Ca	Calcium	
Mg	Magnesium	
Al	Aluminium	
C	Carbon	
Zn	Zinc	
Fe	Iron	
Sn	Tin	
Pb	Lead	
H	Hydrogen	
Cu	Copper	
Ag	Silver	
Au	Gold	
Pt	Platinum	
C	H	Least reactive

added for comparison

Reactivity Series of Metals

Classify each of the following reactions by type: single replacement(SR), double replacement(DR), synthesis(S), decomposition(D), or combustion (C)



Lesson #5: Double Replacement Reactions

Double Replacement Reactions:



1. A and C are _____
2. B and D are _____
3. Double Replacement reactions produce a _____. This is a sign of a chemical reaction.
4. Solubility of a salt depends upon the type of ions in the salt. Some salts are soluble in water and others are not. An _____ solution is a solution with water as the solvent. A compound is said to be _____ if it readily dissolves in water and does not precipitate if left undisturbed for an extended period of time.
5. Ionic compounds that dissolve in water are described as _____

Examples: Sodium chloride, NaCl, and potassium nitrate, KNO₃

When these compounds are mixed with water they dissolve and are aqueous (aq).

6. Ionic compounds that do not dissolve in water are described as _____

Examples: Magnesium hydroxide, Mg(OH)₂

and lead (II) iodide, PbI₂

An insoluble substance simply remains in the solid state (s) when added.

7. You can predict whether a precipitate will form in an aqueous reaction using solubility rules. You do not need to memorize. But you do need to learn how to use them to predict a precipitate.

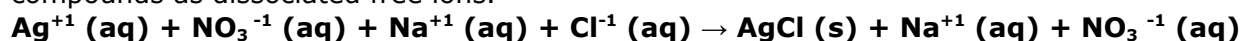
8. Classify the following compounds as soluble or insoluble:



Compounds Containing the Following Ions Are Generally Soluble	Exceptions
Group I and NH ₄ ⁺	none
NO ₃ ⁻ and C ₂ H ₃ O ₂ ⁻	none
Cl ⁻ , Br ⁻ , and I ⁻	when paired with Ag ⁺ , Hg ₂ ²⁺ , or Pb ²⁺ (insoluble)
SO ₄ ²⁻	when paired with Sr ²⁺ , Ba ²⁺ , Pb ²⁺ , Ag ⁺ , or Ca ²⁺ (insoluble)
Compounds Containing the Following Ions Are Generally Insoluble	Exceptions
S ²⁻	when paired with group I, NH ₄ ⁺ , Ca ²⁺ , Sr ²⁺ , or Ba ²⁺ (soluble)
OH ⁻	when paired with group I and NH ₄ ⁺ (soluble) or when paired with Ca ²⁺ , Sr ²⁺ , or Ba ²⁺ (slightly soluble)
CO ₃ ²⁻ and PO ₄ ³⁻	when paired with group I or NH ₄ ⁺

9. Writing the Reaction: Silver nitrate and sodium chloride forms silver chloride and sodium nitrate
Full Reaction: AgNO₃ (aq) + NaCl (aq) → AgCl (s) + NaNO₃ (aq)

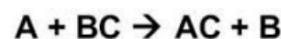
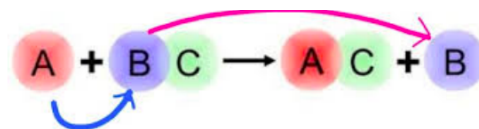
10. A _____ equation is an equation that shows dissolved ionic compounds as dissociated free ions.



11. A _____ ion is an ion that is present during a reaction but does not take part in the reaction. An ion that appears on both sides of an equation and is not directly involved in the reaction is called a spectator ion. **Spectator Ions: Na^{+1} & NO_3^{-1}**
12. A _____ ionic equation is an equation that only shows the ions that undergo changes during a chemical reaction. (Spectator ions are omitted from net ionic equations.) The net ionic equation is an equation for a reaction in solution that shows only those particles that are directly involved in the chemical change. **$\text{Ag}^{+1}(\text{aq}) + \text{Cl}^{-1}(\text{aq}) \rightarrow \text{AgCl}(\text{s})$**

Lesson #6 Single Replacement Reactions

1. Single Replacement Reactions: One element takes the place of another in a compound.



2. A new element and a new compound are produced
- The new _____ that forms is often a solid, but could also be a liquid or a gas
 - The new _____ that forms is aqueous, a solution (which means it is dissociated in the water as ions)
3. Single replacement reactions will only happen in one direction
 $\text{Fe}(\text{s}) + \text{CuSO}_4(\text{aq}) \rightarrow \text{FeSO}_4(\text{aq}) + \text{Cu}(\text{s})$ will occur,
 $\text{Cu}(\text{s}) + \text{FeSO}_4(\text{aq}) \rightarrow \text{CuSO}_4(\text{aq}) + \text{Fe}(\text{s})$ will not occur

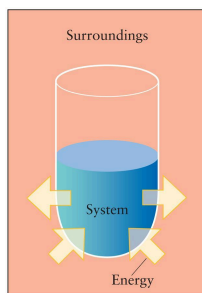
•If a particular single replacement reaction occurs, then the reverse reaction will not occur

4. The reaction that occurs, contains the more reactive metal as the element. Elements that are more reactive, prefer to be in compounds
5. An _____ of metals is a list of metals arranged in order of decreasing chemical activity (i.e., from most to least reactive). It is used to help us predict what chemical reactions will occur.
6. Even though carbon and hydrogen are not metals, they can act as cations so they are often included on the list
7. Practice Using the Activity Series (Write NR if the reaction does not occur)



K	Potassium	↑ most reactive
Na	Sodium	
Ca	Calcium	
Mg	Magnesium	
Al	Aluminium	
C	Carbon	
Zn	Zinc	
Fe	Iron	
Sn	Tin	
Pb	Lead	
H	Hydrogen	↓ least reactive
Cu	Copper	
Ag	Silver	
Au	Gold	
Pt	Platinum	
(added for comparison)		

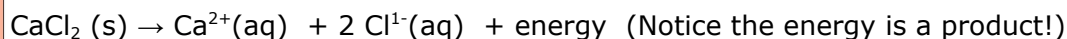
Lesson #7 Exothermic and Endothermic Reactions



System: the mixture of reactants **Surroundings:** everything else!!

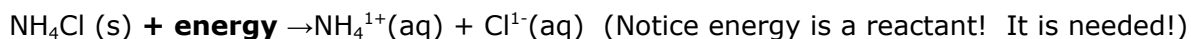
The Law of Conservation of Energy states that energy is neither created nor destroyed in a chemical or physical process.

Exothermic Process: Heat flows from the system into the surroundings
System → Surroundings. Heat is Released! "Exits the system!"



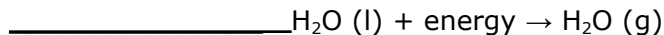
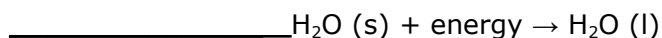
Observations from the video or demonstration:

Endothermic Process: Heat flows from the surroundings to the system.
Surrounding → System. Heat is absorbed or used. "Enters the system!"

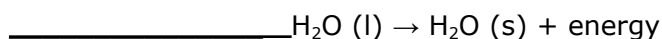


Observations from the video or demonstration:

Endothermic Phase Changes (energy flows into the system)

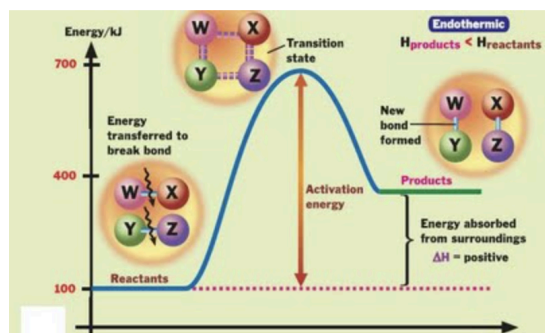


Exothermic Phase Changes (energy must be removed or flow out of the system..cooled)

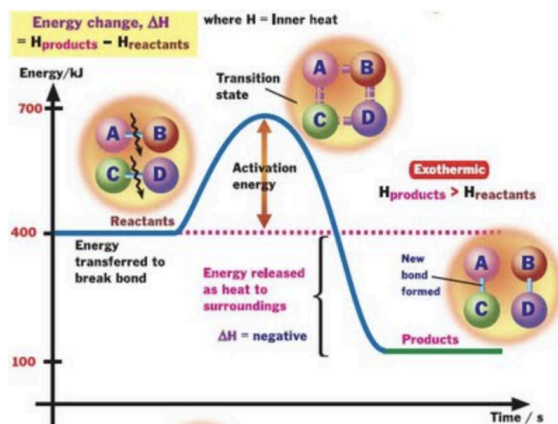


Bond Breaking _____ energy and is _____

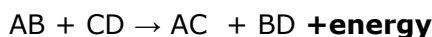
Bond Forming _____ energy and is _____



In an endothermic reaction, bond breaking requires more energy than is released bond formation, so energy must be continually supplied. The energy term is on the reactant side of the equation showing the reaction requires energy to continue.



In an exothermic reaction, bond breaking requires less energy than is released by bond formation, so energy is released. The energy term is on the product side showing that energy is a product of the reaction and is released from the system.



In chemical reactions, bonds store _____ (H) which is also called enthalpy. Energy is measured in _____, _____ or _____.
NOT Celsius or Kelvin!!!

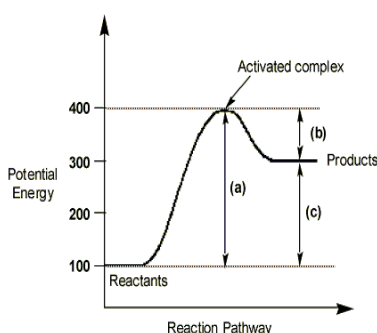
The energy stored by the bonds in the reactants is _____ as the energy stored in the bonds of products.

However, energy is _____ **Energy on Reactant Side = Energy on Product Side**

The change in energy is equal to **$\Delta H = H_{\text{products}} - H_{\text{reactants}}$**

Potential Energy Diagram for an Endothermic Reaction

(System Gains potential energy from the surroundings.!)



Example:

If $H_{\text{products}} > H_{\text{reactants}}$, ΔH is _____. The reaction is _____ and the energy term must be on the _____ side to balance the equation.

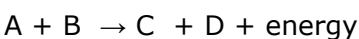
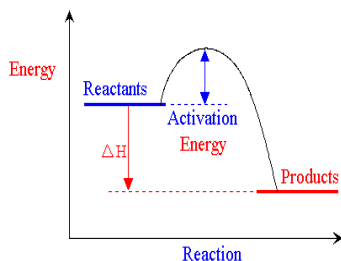


Example Video: $\text{Ba}(\text{OH})_2 + 2\text{NH}_4\text{Cl} + \text{energy} \rightarrow \text{BaCl}_2 + 2\text{NH}_3 + 2\text{H}_2\text{O}$

Observations from the video:

Potential Energy Diagram for an Exothermic Reaction

(System Loses potential energy to the surroundings.)



Example:

If $H_{\text{reactants}} > H_{\text{products}}$, ΔH is _____. The reaction is _____ and the energy term must be on the _____ side to balance the equation.

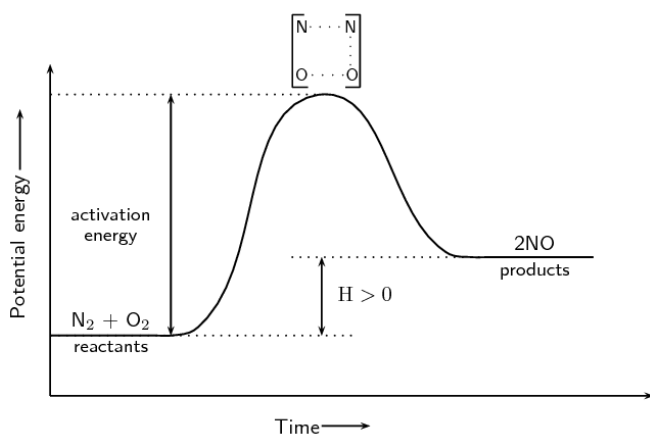
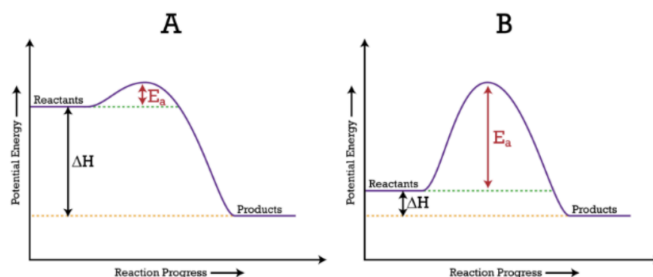


Example Video: $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO} + \text{energy}$

Observations from the video:

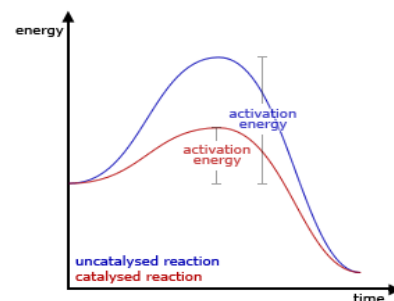
Potential Energy Diagram with Activation Energy and a Catalyst

Activation energy is the energy to break bonds and start a chemical reaction. It is abbreviated _____. A catalyst is a substance that _____ the activation energy and makes the reaction go faster.



An _____ is the transition state molecule, with some bonds partially broken and other bonds partially formed. It is an intermediate state in a reaction. Label the activated complex in the energy diagram.

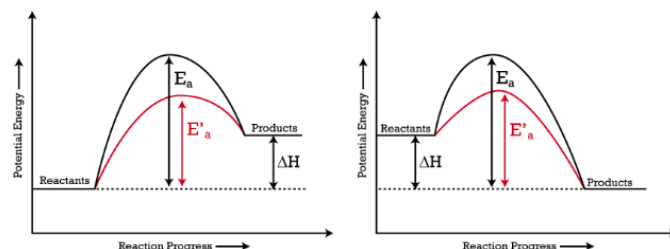
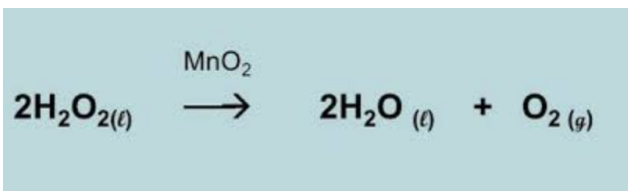
A catalyst is a substance that does not change in a reaction but lowers the activation energy of the reaction and speeds up the rate without changing _____.



Catalysts can be used in both Endothermic and Exothermic Reactions. They change E_a which makes the reaction go faster.

Do they change ΔH ? NO!

Notes on the Video of the decomposition of hydrogen peroxide:



Notes on Fuse School Video: